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Physical Geography of India

Topics covered

- Natural regions and natural vegetation
- The six natural regions
- Rivers of India

TP

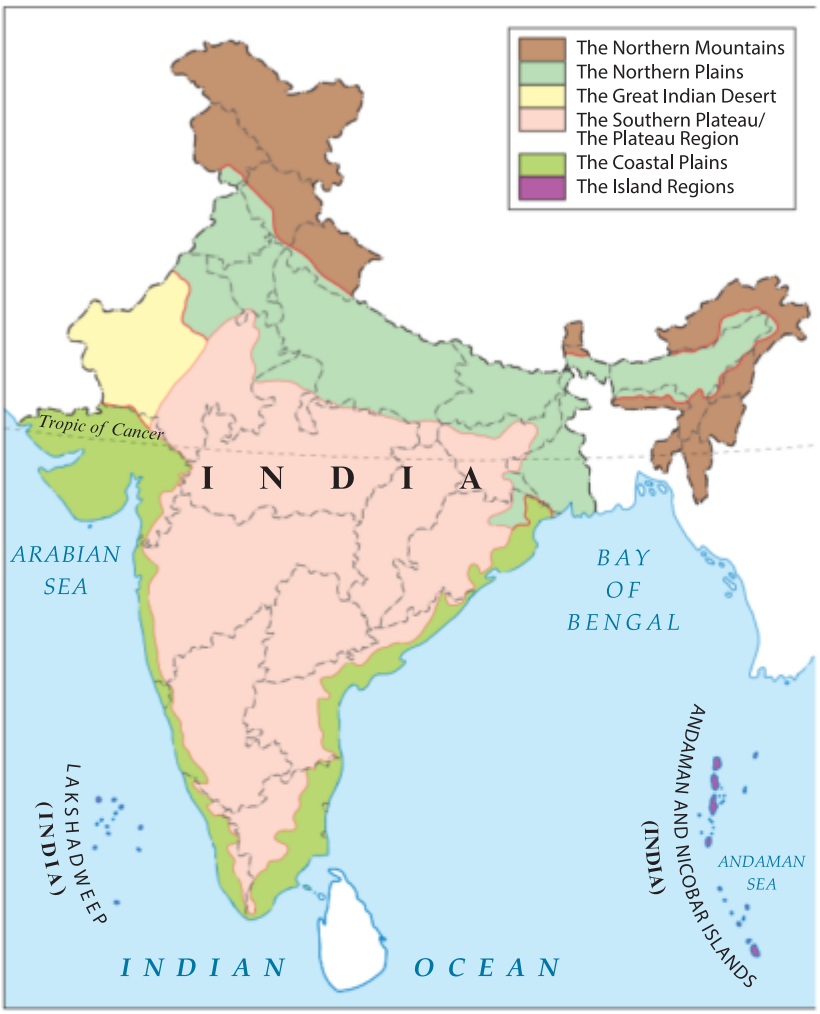
You have learnt that India is in the southern part of Asia. High mountain ranges separate India from a large part of Asia. The land south of the mountain ranges is called the Indian **subcontinent**. In this map, use the legend to find:

- the mountains to the north of the subcontinent
- the plains to the south of the mountains
- two plateaus south of the plains
- the Vindhya and Satpura hill ranges

Physical features of the Indian subcontinent

NATURAL REGIONS AND NATURAL VEGETATION

The Indian subcontinent has many different physical features. India shares these physical features with its neighbours that are also in the subcontinent—Pakistan, Nepal, Bhutan and Bangladesh. The physical features form six different natural regions—regions created by nature. The weather and geography within a natural region are almost similar. These are given in the map here. Match the colours in the legend with the colours in the map to find each region.



Natural regions of India

In each natural region there are certain trees and plants that grow naturally (that are not planted by people). These are called natural **vegetation**. The natural vegetation of an area depends on the kind of land, the kind of weather as well as the height of the land. There are the different kinds of natural vegetation in India.

1. Evergreen rainforests: These grow in areas where there is heavy rainfall. The forests remain green because all the trees do not shed their leaves at the same time.
2. **Deciduous** forests: These are found in areas where there is not that much rain. Deciduous trees shed their leaves every year.
3. Scrub vegetation: This is found in drier places. There are dwarf deciduous trees and grasses.
4. Desert vegetation: This is found in the desert areas. Plants have long roots to search for water and thick stems to store water.
5. Mangrove forests: These grow along the coast. Mangrove trees have roots that can also be seen above the ground.

6. Mountain vegetation: In the lower hills there are forests like bamboo forests. On the higher mountains, there are forests of trees like conifers. These trees have needle-like leaves and woody cones which contain their seeds. The highest slopes mostly have only grass and flowers.

In many parts of India, forests have been cut down for different reasons such as to clear land to grow food.

Needle-like leaves and cones on a conifer in the Himalayas

THE NORTHERN MOUNTAINS AND HILLS OF THE NORTH-EAST

Many ranges of high mountains curve across north India. These are the Himalayas, the highest mountains in the world. They form a natural boundary between India and a large part of Asia. Two neighbouring countries, Nepal and Bhutan, are situated in these mountains.

Geographers have divided the northern mountains according to their heights.

Mt Everest, the highest mountain in the world, is in Nepal.

The Himadri

These are the highest mountains. They are also called the Greater Himalayas. They get heavy snowfall and are always covered in snow. It is very cold here, both in summer and winter. There is no plant or animal life. High mountains like Mt Everest and Mt Kangchenjunga are in the Greater Himalayas.

Forested ranges of the Lesser Himalayas—the snow-covered mountains are also visible.

The Himachal

These mountains are lower than the Himadri and are also called the Lesser Himalayas. There is snow in winter but

summer is cool and pleasant. Some areas have grassy slopes while others have forests.

The Shiwalik

These are the lowest range. It is not as cold here as in the higher mountains. In many places the thick forests of the Shiwalik have been cut down, so the soil gets washed away with heavy rain.

Did you Know?

The Himalayas were created when the subcontinent pushed against Asia. The subcontinent is still moving slowly towards Asia and so the Himalayas are rising between 1-2 centimetres each year!

Hills of North-east India

In eastern India, the mountain ranges turn southwards and become hill ranges. They are covered with forests and greenery as it rains a lot here. The hills have different names like the Mizo Hills or the Garo Hills.

The Effect of the Himalayas

- The area north of the Himalayas is cold and windy. The high mountains prevent cold winds from this region from blowing over India. This makes India much warmer than places that are north of the Himalayas.
- The Himalayas help rain to fall on the Indian subcontinent. When rain-bearing clouds from the seas reach the mountains, they cannot go any further. They rise up into the cold, upper atmosphere. Here the water vapour turns to rain or snow, which falls on the subcontinent.
- Snow from the mountains melts in summer and fills many of the rivers. These rivers have water even in the hottest months.

Think and answer

SI What would the weather in the Indian subcontinent be like if there were no high mountain ranges in the north? (Hint: winds, monsoon)

THE NORTHERN PLAINS

The Northern Plains are located to the south of the Himalayas. In the west they extend into Pakistan. Bangladesh is situated on the eastern part of the plains. Many huge

rivers flow through the flat land of the plains. These rivers carry **silt** (mixture of sand and mud) as well as **gravel** (small stones). In the monsoon, the rivers often flood. When this happens, the silt and gravel spread over the land. This is called **alluvium**. This process of flooding, and alluvium spreading over the land, has helped to form the northern plains over thousands of years. Plants grow very well in alluvium.

THE GREAT INDIAN DESERT

The western part of India is a dry desert that gets very little or no rain, year after year. Some areas are very sandy, with sand dunes, whereas in other parts there is desert vegetation. This desert is called the Thar Desert and it extends into Pakistan in the west.

THE SOUTHERN PLATEAU

The plateau region lies to the south of the Great Northern Plains and east of the dry western areas. It has different names in different parts. The southern, peninsular part is called the Deccan Plateau. The Deccan Plateau is almost completely surrounded on three sides by rocky hills. These are the Vindhya and Satpura ranges in the north, the Western Ghats in the west and the Eastern Ghats in the east.

A hill at Bandhavgarh in Madhya Pradesh with a flat top, like a table

The vast plateau region is not the same all over. Some parts are quite low, while others are higher. In some parts there are huge rocks. In many parts of the plateau region, there are forests with deciduous trees.

THE COASTAL PLAINS

The coastal plains are narrow **strips** of flat land that form a V shape around the plateau region. The eastern coastal plain lies between the Bay of Bengal and the Eastern Ghats. The western coastal plain lies between the Arabian Sea and the Western Ghats. The eastern coastal plain is wider than the western one. The two plains join at the southern tip of peninsular India. Here you can see the sun rising and setting over the sea.

Sunrise over the sea

The coastal plains have lovely beaches with tall coconut trees along the coast. The coconut trees, mangrove forests and other trees growing along the coast protect the land from storms and cyclones.

Think and answer

SI

- Why is it possible to see the sunrise and the sunset over the sea at the southern tip of the peninsula?
- If you stand on the beach here, looking towards Sri Lanka in the south, which hand will point towards the rising sun?

THE ISLAND REGIONS

The island regions of India are two archipelagos on either side of peninsular India. The Lakshadweep Islands are in the Arabian Sea and the Andaman and Nicobar Islands are in the Bay of Bengal.

The Lakshadweep Islands have 36 coral islands. Coral is made up of the hard skeletons of millions of tiny sea animals that pile up over thousands of years until they rise above sea level. The action of the wind and waves breaks down the coral into sand. Bird droppings make the sand fertile and so plants are able to grow.

Coconut trees along a sandy beach on the eastern coast

White sand and very flat land of a coral island

The Andaman and Nicobar Islands are a string of over 550 islands. It rains a lot in these very hot islands. The heavy rains are good for plants and so there are thick forests here.

Discuss

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Have you visited any places in these natural regions? Share your experiences in class.

THE RIVERS OF INDIA

The Indian subcontinent has many rivers. In the higher mountains and hills, these rivers flow very fast. The flowing water carries with it silt and stones. In the plains, the rivers flow more slowly. They carry the silt and stones to the mouth. If the river floods, they spread over the land.

India's rivers are either perennial or non-perennial. Look at the map as you read about these rivers.

The huge Brahmaputra River in Assam

Perennial rivers

These have water all the year round. All these rivers are in north or north-east India. They have their source in the Himalayas, or in snowy regions beyond the Himalayas. They get water from melting snow, so are also called snow-fed rivers. Apart from water from melting snow, they can also get very full of water if the monsoon rain is heavy. When this happens, the rivers flood the surrounding areas.

In the hills and in the plains other, smaller, rivers join the main rivers. These are called **tributaries**. Some examples of the main perennial rivers are the Indus, Ganga and Brahmaputra. The Satlej, Yamuna and Tista are examples of snow-fed tributaries.

Some snow-fed and rain-fed rivers in India

Non-perennial rivers

Non-perennial rivers start because of heavy rain in the hills, so are also

called rain-fed rivers. As they are rain-fed, they sometimes do not have any water when there is less rainfall. Rain-fed rivers are found in central and south India. In the Deccan Plateau many rivers start in the Western Ghats and empty into the Bay of Bengal. They flow eastwards as the plateau is higher in the west. Two such rivers are the Mahanadi and Krishna. The Narmada and Tapi start in the plateau region, flow west and empty into the Arabian Sea. Some rivers, like the Betwa and Son, start in the northern part of the plateau and flow towards the Great Northern Plains.

Write here the names of two perennial and two non-perennial rivers:

.....

.....

Rivers are very useful. For example:

- People use their water for growing crops.
- In many places, river water is used to produce electricity.

People in all the areas through which these rivers flow need to cooperate with each other so that everyone is able to benefit from them. The rivers of India, and its geography, link the people of the different regions of India.

Think and answer

SI

If people living near the upper parts of a river use up most of its water, how will this affect the people who live near the lower parts of the river?

Values and Life Skills

Why should we learn about the various regions and people of India?

REMEMBER

1. The Indian subcontinent has many different physical features. India shares these with the other countries of the subcontinent.
2. India is divided into six natural regions based on the physical features of the land.
3. There are similarities in the weather and geography within a natural region.
4. Each region has its own type of natural vegetation.
5. India has many perennial and non-perennial rivers.
6. The physical features and rivers link the people of India.

NEW WORDS

Subcontinent	a large landmass that is part of a larger continent
Vegetation	plants, especially the plants that are found in a particular area or environment
Deciduous	a tree that loses its leaves every year
Silt	sand and mud that is carried by flowing water
Gravel	small stones
Alluvium	silt and gravel that are left by rivers or floods
Strip	a long and narrow area of land
Tributary	a stream or river that flows into a larger river

EXERCISES

A. Match the following.

A	B
1. scrub vegetation	a. heavy rainfall, forests remain green
2. mangrove forests	b. not that much rain, trees shed their leaves every year
3. mountain vegetation	c. drier places, dwarf deciduous trees and grasses
4. evergreen rainforests	d. deserts, plants with long roots and thick stems
5. deciduous forests	e. coasts, trees with roots seen above the ground
6. desert vegetation	f. hills and mountains, bamboos, conifers and grass

B. Crossword

Write the clues for this crossword in your notebook.

C. Complete the sentences in your notebook.

1. The hills of the north-east are covered with ..... .
2. Three plateaus in the plateau region are ..... .
3. The two archipelagos of India are ..... .
4. Perennial rivers are rivers that ..... .
5. Non-perennial rivers are rivers that ..... .

D. Write true or false.

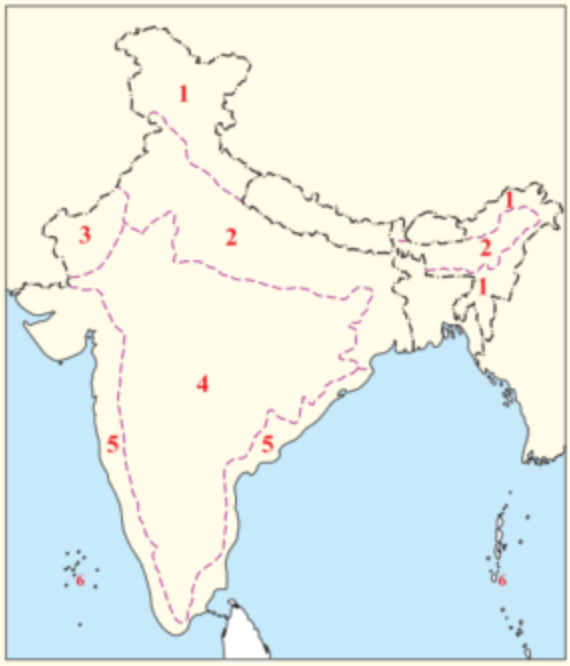
Correct the false sentences in your notebook.

1. The highest mountains are cold and snow-covered, but crops can be grown here.
2. The desert region gets very little rainfall, year after year.
3. The plateau region is not the same all over.
4. The eastern coastal plain is narrower than the western coastal plain.
5. The island regions are both in the Arabian Sea.

E. Answer the following questions.

1. Write two or three sentences to explain what natural vegetation means.
2. What are the three divisions of the Himalayas?
3. Name the three ways in which the Himalayas are useful.
4. Write four sentences to describe the Northern Plains.
5. Which are the hill ranges that surround the Southern Plateau?
6. How are coral islands formed?
7. Why do the Andaman and Nicobar Islands have thick forests?
8. Name two ways in which rivers are useful.

TP THINGS TO DO



A. Picture study

Write the names of the numbered natural regions in your notebook.

SI B. On your own

Find out where these rivers have their source, which natural regions they flow through and where they finish their journey.

- Ganga
- Brahmaputra
- Yamuna
- Satluj
- Krishna
- Betwa
- Narmada
- Kaveri
- Chambal

SI C. In a group

Make groups of three or four. Choose a natural region. Find out what birds and animals are found in the region. Present your research in class.

WEBLINK

India's natural regions: https://www.youtube.com/watch?v=mj8wLzegfFk&list=PLyguaZzohEdYHB5l98yhrqNj3FiZLeUuC (accessed on 27 July 2017)